3.4 Areas That Prove Inequalities

WORKSHEET

The question

STEP III 2011, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q4

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself $\int_0^a \mathrm{f}(x)\,\mathrm{d}x$ is the area between the curve and the x-axis. What area does $\int_0^b \mathrm{f}^{-1}(y)\,\mathrm{d}y$ represent on the same picture?

STAGE 1 The picture

Covers: Part (i)

Ask yourself $\int_0^a \mathrm{f}(x)\,\mathrm{d}x$ is the area between the curve and the x-axis. What area does $\int_0^b \mathrm{f}^{-1}(y)\,\mathrm{d}y$ represent on the same picture?

Pause and try Draw the rectangle with corners $(0,0)$ and (a,b), and shade both regions. Do they cover the rectangle? When is there no excess?

STAGE 2 Young's inequality

Covers: Part (ii)

Ask yourself With $f(x) = x^{p-1}$, what is $f^{-1}(y)$? Work out both integrals.

Pause and try Use $\frac{1}{p} + \frac{1}{q} = 1$ to simplify the second integral. For equality, use your answer to part (i).

STAGE 3 An inverse sine

Covers: Part (iii)

Ask yourself Take $f(x) = \sin x$ on $[0, \frac{\pi}{2}]$. How do you integrate $\arcsin y$? Try integration by parts, with $\arcsin y \times 1$.

Pause and try To deduce the inequality for $\arcsin(t^{-1})$, choose the simplest possible value of a, and put $b = \frac{1}{t}$.

What's really going on

From Young to Hölder

William Henry Young published his inequality in 1912. Its importance comes from what it leads to. Suppose $\sum a_i^p = 1$ and $\sum b_i^q = 1$. Applying Young to each pair and adding,

$$\sum a_i b_i \le \frac{\sum a_i^p}{p} + \frac{\sum b_i^q}{q} = \frac{1}{p} + \frac{1}{q} = 1.$$

Scaling the as and bs to make the sums 1 gives *Hölder's inequality*: for positive numbers,

$$\sum a_i b_i \le \left(\sum a_i^p\right)^{1/p} \left(\sum b_i^q\right)^{1/q}.$$

With $p = q = 2$ this is the *Cauchy–Schwarz inequality*, $\sum a_i b_i \le \sqrt{\sum a_i^2}\sqrt{\sum b_i^2}$, which says that the cosine of the angle between two vectors is at most 1. Hölder's inequality

is the foundation of the theory of L^p spaces, spaces of functions in which "distance" is measured by $(\int |f|^p)^{1/p}$, and which you will meet in courses on functional analysis and probability.

The Legendre transform

The picture in Figure 3.4.1 (in the full section online) treats f and f^{-1} completely symmetrically. Write $F(a) = \int_0^a \mathrm{f}$ and $G(b) = \int_0^b \mathrm{f}^{-1}$. The inequality says $F(a) + G(b) \geq ab$, with equality on the curve, so

$$G(b) = \max_a (ab - F(a)).$$

This operation, turning a convex function F into a new convex function G, is the *Legendre transform*, and applying it twice gets you back to F. In mechanics it converts the *Lagrangian*, a function of position and velocity, into the *Hamiltonian*, a function of position and momentum: two equivalent formulations of Newton's laws that you will meet in a second-year course. In thermodynamics it converts between energy, enthalpy and free energy. For $F(a) = a^p/p$ the transform is $G(b) = b^q/q$, which is why the two exponents in Young's inequality are linked by $\frac{1}{p} + \frac{1}{q} = 1$.

Try this next

1. **The 1987 version.** Try STEP I 1987, Question 6 (step.maths.org/questions/87-s1-q6). It uses the same kind of picture to integrate an inverse function. Check your answer by differentiating.

2. **Cauchy–Schwarz.** Use Young's inequality with $p = q = 2$ to prove $a_1b_1 + a_2b_2 + a_3b_3 \leq \sqrt{a_1^2 + a_2^2 + a_3^2}\sqrt{b_1^2 + b_2^2 + b_3^2}$ for positive numbers. (*Hint:* first deal with the case where both square roots equal 1.)

3. **An exponential version.** Take $\mathrm{f}(x) = e^x - 1$ in $(*)$ to show that, for $a, b \geq 0$,

$$ab \leq e^a - 1 - a + (1 + b)\ln(1 + b) - b.$$

   When is there equality?

4. **How good is the bound?** HARDERUse a calculator to compare $\arcsin(1/t)$ with $t - \sqrt{t^2 - 1}$ for $t = 1, 2, 10$. Show that for large t, $t - \sqrt{t^2 - 1} \approx \frac{1}{2t}$ while $\arcsin(1/t) \approx \frac{1}{t}$. Why is the bound off by a factor of 2? (Look at the picture with $a = 0$: how much of the excess is wasted?)

Further reading

- **J. Michael Steele, *The Cauchy–Schwarz Master Class*** (Cambridge University Press). Chapters 1 and 9 cover Cauchy–Schwarz and Hölder, with Young's inequality along the way.

- **Roger Nelsen, *Proofs Without Words*** (Mathematical Association of America). Includes Young's inequality and many other picture proofs.
- **V. I. Arnold, *Mathematical Methods of Classical Mechanics*** (Springer). Section 14 explains the Legendre transform with exactly this picture. Advanced, but the relevant section is short.

The full guided solution is in Section 3.4:
https://beyondstep.keithmcnulty.org/ch03/areas-inequalities.html